

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Mukesh Raj L(192571036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Task 1: Article Summary

1.1 Formal Bibliographic Citation

Guibin Lv, Fudong Dong, Pengyuan Li, Yuansen Wang, Yongguo Sheng, Feng Cheng, and Bo Wang, "Highway Engineering Construction Safety Risk Evaluation Based on Decision Tree Algorithm," *International Journal of Computational Intelligence Systems*, vol. 19, no. 166, pp. 1–20, 2026. DOI: <https://doi.org/10.1007/s44196-026-01207-0>

1.2 Domain and Machine Learning Problem Classification

The domain of this research is highway infrastructure construction engineering, specifically focusing on safety risk management and early warning within complex highway projects (such as bridges, elevated roads, and roadbed construction). In civil engineering, safety management is historically characterized by qualitative inspections, high uncertainty, and complex interactions between diverse factors including human operations, management overheads, mechanical conditions, and unexpected environmental conditions.

From a computer science and machine learning perspective, the paper addresses a supervised binary classification and feature-weighting problem. The classification objective is to predict whether a specific combination of safety risk indicators will lead to a safety incident during construction (the target variable is 'Whether it leads to safety incidents in construction', taking binary labels of 'YES' or 'NO'). Concurrently, the algorithm addresses a feature-weighting problem where the conditional features represent the frequency of quaternary risk factors. The information gain metric from the inductive decision tree is utilized to calculate objective, data-driven weight coefficients for each risk factor.

1.3 Objectives and Core Research Gaps

The key objective of this research is to establish a digital safety risk evaluation model that integrates the interpretability of inductive machine learning with the quantitative structure of engineering risk matrices. By overcoming the subjectivity of traditional weighting schemes, the model aims to provide site managers with a reliable, dynamic, and highly visual early-warning tool to identify and mitigate. The authors identify several major research gaps in traditional civil engineering risk assessment approaches:

Subjectivity of Traditional Weighting: Conventional methods such as the Analytic Hierarchy Process (AHP) and the G1 method rely on subjective expert judgments and pairwise comparisons, which can lead to significant human bias and weight deviations.

Causal Logic Opacity: Objective weighting methods like the entropy weight method, while mathematically unbiased, are unable to represent the actual impact logic or decision pathway mapping how combinations of risk factors trigger safety accidents.

Inability to Model Coupling and Interdependencies: Traditional models often treat risk factors as completely independent, failing to address multi-factor coupling and nonlinear correlation where minor, co-occurring risk factors combine to trigger catastrophic events.

Static and Inflexible Frameworks: Many existing frameworks rely on static parameters, making them unable to adapt to dynamically changing construction conditions, such as sudden weather shifts or sudden drops in mechanical maintenance quality.

To fill these critical gaps, the research introduces the ID3 (Interactive Dichotomiser 3) decision tree algorithm into the risk calculation flow. Rather than predicting accidents as a black box, the ID3 algorithm is used to analyze historical frequency datasets and generate a clear, logical decision tree. The information gain calculated during node splitting is normalized to serve as objective weight coefficients for each risk factor. These weights are then integrated into a dual-dimensional risk matrix (mapping probability and consequence severity) to construct a dynamic, highly interpretive safety risk index. This hybrid framework balances the mathematical objectivity of data-driven models with the qualitative wisdom of field engineering standards.

Task 2: Methodology Analysis

2.1 Technical Exposition of Machine Learning Methods

The research framework develops a hybrid model combining the ID3 decision tree algorithm and a fuzzy risk matrix. The methodology operates in two distinct phases: First, data-driven weight derivation using the ID3 decision tree; second, the quantitative calculation of risk consequence severity utilizing a fuzzy membership function combined with the G1 weighting scheme.

2.1.1 Data-Driven Feature Weighting via the ID3 Algorithm

The ID3 algorithm is an inductive supervised learning method used to construct a decision tree by selecting the most informative attributes through information entropy and information gain. In this research model, the conditional attributes represent the frequencies or occurrences of different construction safety risk factors, while the decision attribute indicates whether a construction safety incident occurred, with two possible classes: YES and NO. The first step in the ID3 algorithm is calculating the information entropy of the decision attribute, denoted as $\text{Info}(D)$, which represents the overall uncertainty in predicting safety incidents. It is calculated using the formula $\text{Info}(D) = -\sum p_j \log_2(p_j)$, where p_j represents the probability of the j th class. Since there are two classes, YES and NO, the formula becomes $\text{Info}(D) = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$, where p_1 is the probability of an accident occurring and p_2 is the probability of no accident occurring.

The next step is calculating the conditional entropy, represented as $\text{Info}_x(D) = \text{Info}(D|X) = \sum p_i \times \text{Info}(X_i)$, which measures the uncertainty remaining after the dataset is divided according to a particular risk factor X . Here, p_i represents the probability of the i th value of the risk factor, while $\text{Info}(X_i)$ represents the entropy of the corresponding subset. The information gain of each risk factor is then calculated using $\text{Gain}(X) = \text{Info}(D) - \text{Info}_x(D)$. A higher information gain indicates that the risk factor has greater predictive ability in determining whether a construction safety incident may occur. Finally, the information gain values are converted into normalized weight coefficients using $w_i = \text{Gain}(X_i) / \sum \text{Gain}(X_i)$. These normalized weights represent the relative importance of each risk factor, and their total sum is equal to one. Therefore, the ID3 algorithm provides an objective and data-driven approach for determining the importance of different construction safety risk factors.

2.1.2 Fuzzy Membership Models for Risk Consequence Severity

To calculate the consequence severity of each risk factor, a fuzzy membership model is established to evaluate its impact on three important construction asset categories: Personnel, Third-Party Assets, and Operational Equipment. The expert evaluation scores for these three categories are represented as i_1 , i_2 , and i_3 , respectively. These scores are transformed into fuzzy membership degrees across five consequence severity levels, namely Level I (Very Minor), Level II (Minor), Level III (Moderate), Level IV (Severe), and Level V (Catastrophic). Fuzzy membership functions are used because the severity of a risk may not always belong completely to a single level and can have partial membership in adjacent severity levels.

For Level I, representing Very Minor consequences, the membership function is defined as $\mu_{tI}=1$ when $it \geq 4$, and $\mu_{tI}=0$ when $0 \leq it < 4$. For Level II, representing Minor consequences, the membership value is $\mu_{tII}=0$ when $it \geq 4$, $\mu_{tII}=it-3$ when $3 \leq it < 4$, and $\mu_{tII}=0$ when $0 \leq it < 3$. This membership function determines the degree to which the evaluated consequence belongs to the Level II category.

For Level III, representing Moderate consequences, the membership function is defined as $\mu_{tIII}=0$ when $it \geq 4$, $\mu_{tIII}=4-it$ when $3 \leq it < 4$, $\mu_{tIII}=it-2$ when $2 \leq it < 3$, and $\mu_{tIII}=0$ when $0 \leq it < 2$. This level represents a

moderate degree of impact and allows the score to gradually transition between neighbouring severity levels.

For Level IV, representing Severe consequences, the membership function is given by $\mu_{IV}=0$ when $it \geq 3$, $\mu_{IV}=3-it$ when $2 \leq it < 3$, $\mu_{IV}=it-1$ when $1 \leq it < 2$, and $\mu_{IV}=0$ when $0 \leq it < 1$. This function represents risks with a high level of potential consequence.

Finally, Level V represents Catastrophic consequences, which correspond to the most serious level of risk impact. Its membership function is defined as $\mu_V=0$ when $it \geq 2$, $\mu_V=2-it$ when $1 \leq it < 2$, and $\mu_V=1$ when $0 \leq it < 1$. By calculating the membership values for all five levels for each of the three asset categories, the model captures the uncertainty and gradual variation in consequence severity. These membership values are subsequently used to construct the 3×5 fuzzy evaluation matrix M , which forms the basis for the comprehensive fuzzy evaluation and the calculation of the final risk consequence severity score.

2.2 Dataset Architecture and Preparation

The empirical data supporting this methodology originates from the construction monitoring logs of a major infrastructure project: the Second Phase of a city's Inner Ring Expressway Speed-up Transformation Project. Spanning 5.16 kilometers with a red-line width of 60 meters, this project presents high technical difficulty due to its complex construction environment involving elevated mainline structures, ground-level pavements, an interchange point, and multiple ramps.

Over an observation period spanning from 2016 to 2021, the authors collected 12 semi-annual records. Each semi-annual record documents the precise frequency of occurrence of each quaternary safety risk indicator alongside the occurrence of construction safety accidents (YES or NO). In total, the database comprises 3,726 individual risk events across the entire observation period. For the personnel category, the dataset contains a subset of 552 sample events. This historical frequency data.

Data Preparation and Discretization: Decision tree algorithms like ID3 require discrete features to calculate information gain effectively. Raw frequencies are naturally continuous and highly variable, which would lead to an excessively wide, shallow tree and severe overfitting if left untreated. To resolve this, the authors categorized the continuous frequency values into three discrete ordinal groups based on risk management standards and empirical engineering boundaries: '0–9' (Low/Infrequent), '10–19' (Moderate/Frequent), and ' ≥ 20 ' (High/Very Frequent).

2.3 Mapping to Course Outcome 4 (CO4) Syllabus

Inductive Learning and ID3 Algorithm: The paper presents a textbook application of inductive learning, specifically the Interactive Dichotomiser 3 (ID3) algorithm. Students analyze how a system learns general classification rules (identifying what combines to cause an accident) from a finite set of historical training instances.

Information Theory Concepts: The mathematical derivations of Information Entropy and Information Gain directly reinforce the theoretical concepts of entropy as a measure of dataset impurity and information gain as the criterion for optimal feature selection.

Decision-Making and Rule Extraction: The paper demonstrates how complex, multidimensional, and non-linear civil engineering risks are transformed into highly legible logical rules (e.g., IF X_{123} in [10-19] AND X_{121} in [0-9] THEN NO ACCIDENT). This aligns with the CO4 goal of building models that support intuitive and transparent corporate decision-making.

2.4 Systematic Methodological Critique: Strengths and Limitations

A rigorous analysis of the authors' methodology reveals several distinct strengths and critical limitations:

Methodology strength: The primary strength of the proposed model is its ability to overcome the

subjective bias of traditional civil engineering weighting systems. By using the information gain of the ID3 decision tree, the weights are derived entirely from empirical, historical project data. This ensures that features that are mathematically proven to reduce the uncertainty of accident outcomes are assigned higher weights. This data-driven objectivity provides a highly defensible weighting framework compared to AHP, where the subjective opinions of a small panel of experts are prone to cognitive biases, fatigue, and internal inconsistencies.

Methodological Limitation: Assumption of Feature Independence during Splitting

The core limitation of the authors' approach lies in the ID3 algorithm's fundamental assumption of feature independence during attribute selection. In complex civil engineering scenarios, risks are highly coupled and interdependent (e.g., extreme weather X_41 directly accelerates equipment deterioration X_332, and poor technical disclosure X_231 compound human error X_122). Because ID3 evaluates information gain for each feature individually, it fails to model these dynamic causal correlations. Consequently, the derived weights and generated decision rules may misrepresent the systemic risk profile when multiple, highly correlated risk factors are present simultaneously.

Task 3: Results & Critical Evaluation

3.1 Synthesis of Case Study Results

By applying the ID3 decision tree-risk matrix model to the city's Inner Ring Speed-up Transformation Project, the authors calculated the safety risk values for all risk factors. The final compiled safety risk index for the entire construction project was computed to be 8.96. According to the risk assessment matrix threshold standards, an index value between 5 and 9 is classified as a Moderate/Medium Risk level.

The analysis of secondary risk categories reveals the primary drivers of this risk level:

Management Risk (X3): Had the highest weight coefficient of 0.288 and the highest calculated risk value of 9.93, indicating that organizational and managerial factors are the most significant contributors to safety risks in this project.

Technical Risk (X2): Followed closely with a weight coefficient of 0.266 and a risk value of 9.33, highlighting the impact of complex technical measures, technical disclosure, and engineering plans.

Environmental Risk (X4): Had a weight coefficient of 0.212 and a risk value of 8.42, capturing the impact of seasonal extreme weather and geological conditions on the site.

Personnel Risk (X1): Had a weight coefficient of 0.234 and a risk value of 7.83, reflecting the influence of professional qualifications and physical/cognitive states of the labor force.

The high risk values calculated for Management Risk and Technical Risk align with the practical realities of the Inner Ring project, which is situated in a high-density urban corridor requiring complex elevated mainline construction, heavy mechanical equipment, and strict coordination to prevent traffic disruptions. The overall Moderate Risk classification indicates that while the project is manageable, immediate preventive measures, such as establishing dedicated safety supervision agencies and conducting regular equipment audits, are required to prevent risks from escalating to critical thresholds.

3.2 Benchmark Model Comparative Analysis

To validate the performance and unique positioning of the proposed **ID3 Decision Tree–Risk Matrix model**, the proposed methodology was compared with three traditional evaluation methods. The comparative results are presented in Table 1.

Evaluation Method	Personnel Risk Weight (X1)	Technical Risk Weight (X2)	Management Risk Weight (X3)	Environmental Risk Weight (X4)	Calculated Risk Level	Model Focus & Alignment
Proposed Decision Tree–Risk Matrix	0.234	0.266	0.288	0.234	Moderate Risk	Management quality and technical complexity
Combined Weighting–2D Cloud Model	0.439	0.188	0.219	0.155	Low Risk	Over-emphasizes personnel qualifications
Improved Entropy Weight–Matter-Element	0.277	0.260	0.248	0.215	Low Risk	Smoothed and balanced weight distribution
AHP–Fuzzy Comprehensive Evaluation	0.232	0.279	0.333	0.156	Moderate Risk	Heavy reliance on expert opinion weights

Table 1: Comparative Analysis of the Proposed Model and Benchmark Evaluation Methods

Critical Insight from Comparative Data: The comparative analysis reveals that the proposed Decision Tree–Risk Matrix method and the AHP–Fuzzy Comprehensive Evaluation method both classify the project as 'Moderate Risk', while the other two benchmarked models evaluate it as 'Low Risk'. The discrepancies lie in how the weights are allocated across the categories. The Combined Weighting–2D Cloud Model assigns an exceptionally high weight to Personnel Risk (0.439), which leads to an underestimation of risk because the personnel on this highly controlled urban project were well-qualified. Conversely, the proposed Decision Tree–Risk Matrix method correctly identifies that Management Risk (0.288) and Technical Risk (0.266) are the true systemic vulnerabilities of this highly complex construction project, yielding a more realistic and conservative 'Moderate Risk' assessment.

3.3 Critical Evaluation of Metrics and Potential Biases

While the results demonstrate the model's practicality, a deep critique reveals potential biases

1. Delphi Method Coefficient of Variation (< 0.3): The authors state that they used the Delphi method with multiple rounds of anonymous expert scoring to evaluate probability indices, continuing until the coefficient of variation (CV) fell below 0.3. While a $CV < 0.3$ is a standard threshold to show statistical consensus among experts, it does not guarantee the objective accuracy of the scores themselves. If the expert panel shares similar professional backgrounds, they may achieve consensus (low CV) while still being collectively biased. This is a common pitfall of expert-based evaluation systems.

2. Single-Project Dataset Bias: The historical frequency dataset is drawn entirely from a single urban infrastructure project in a specific city (the Inner Ring project) from 2016 to 2021. This localized dataset presents significant geographical and operational biases. For instance, the environmental factors (X4) and management structures (X3) of a highly regulated urban project are very different from those of high-altitude highway tunnels or remote mountainous roads. Applying a decision tree

trained on this urban dataset to other geological regions would lead to poor generalization

3. Lack of Cross-Validation: The paper lacks standard machine learning validation protocols. To establish true model reliability, the authors should have split the historical event data into training and testing subsets, using metrics such as classification accuracy, precision, recall, and F1-score to evaluate the decision tree's predictive power. Without rigorous cross-validation, there is a substantial risk that the decision tree's rules are overfitted to the specific historical events of the Inner Ring project, reducing its robustness for prospective risk forecasting.

3.4 Real-World Applicability and Operational Scaling Challenges

Deployability: The ID3-Risk Matrix model is highly suitable for on-site deployment in major civil engineering projects such as highway roadbed construction, bridge and culvert work, and tunnel excavation, where safety records are systematically logged. It can be integrated into digital project management systems (such as BIM or digital twins) to provide a dashboard visualizing active risk. However, scaling this solution to industry-wide application presents several formidable challenges:

Data Collection and Sparsity Bottlenecks: The model requires large, high-quality historical datasets documenting the frequency of specific risk factors and corresponding accident outcomes. In practice, many construction companies do not log minor safety vulnerabilities (e.g., minor equipment delays or temporary personnel health issues) with the level of detail required to train a reliable decision tree. This data sparsity makes it difficult to train localized models.

Model Drift and Maintenance: As construction technologies, safety standards, and regional environmental conditions change, the historical decision tree rules will become obsolete (model drift). Maintaining the model requires continuous data collection, regular retraining, and expert oversight, which demands specialized data science skills that are often unavailable in civil engineering firms.

Ethical and Liability Implications: Relying on automated machine learning models to determine safety risk thresholds introduces severe ethical and liability concerns. If a decision tree predicts a 'Low Risk' level for a specific set of parameters, and an accident subsequently occurs, where does the liability lie? Civil engineering safety regulations must define how automated risk metrics are integrated into the chain of command, ensuring they supplement, rather than replace, human professional accountability.

Task 4: Reflection & Course Integration

4.1 Deepening Course Understanding (CO4 Integration)

Reading and critically analyzing this article has provided three major insights that deepen my understanding of Course Outcome 4 (CO4) within the CSA17 Artificial Intelligence syllabus:

1. **Real-World Application of Inductive Bias and Occam's Razor:** In the classroom, the choice of splitting criteria in decision trees is often treated as a purely mathematical exercise. This paper demonstrates the practical importance of inductive bias. By preferring shorter trees with high information gain nodes at the root (such as 'Low level of professional skills X_123'), the model naturally focuses managerial attention on the most critical factors. This directly illustrates Occam's Razor—building the simplest logical model that explains the data to maximize clarity and
2. **The Synergy of Soft Computing and Crisp Machine Learning:** The integration of the fuzzy membership functions (for consequence severity) and the discrete ID3 decision tree (for risk probability weighting) highlights how soft computing and symbolic machine learning can work together. In civil engineering, data is often vague and qualitative. Translating expert scores into fuzzy sets before multiplying them by data-driven weights shows how machine learning can adapt to real-world datasets that are otherwise too noisy for standard crisp algorithms.
3. **Interpretability as a Core Requirement of Safety-Critical AI:** In deep learning and reinforcement learning, models are often treated as 'black boxes' where decision-making logic is hidden behind thousands of parameters. This article demonstrates why decision trees are still highly valuable in safety-critical domains like civil engineering. The ability to translate trained tree paths into explicit, visual IF-THEN rules allows safety managers to trace exactly why a specific site state is classified as high-risk, providing the transparency required to justify stopping work or implementing expensive mitigations.

4.2 Proposed Research Extension: A Hybrid ID3-Bayesian Network Model

To address the major limitation identified in the methodological critique—specifically, the ID3 algorithm's inability to model correlations and interdependencies among risk factors—I propose a novel research extension: the development of a **Hybrid ID3-Bayesian Network (ID3-BN) Safety Risk conceptual framework and methodology**.

In this hybrid model, the ID3 decision tree is first used to perform attribute selection and filter out low-impact risk indicators, reducing the dimensionality of the risk indicator system. The remaining high-impact risk factors are then used to construct a Bayesian Network (BN). Unlike decision trees, a Bayesian Network is a directed acyclic graph (DAG) that represents joint probability distributions over a set of variables, making it exceptionally suited for modeling causal dependencies and conditional probabilities between risk factors (e.g., how extreme weather X_41 influences equipment failure X_332, which in turn influences human error X_122).

4.2.1 Mathematical Formulation of the ID3-BN Model

The proposed ID3-BN model integrates the feature selection capability of the ID3 Decision Tree algorithm with the probabilistic reasoning capability of a Bayesian Network (BN). The mathematical formulation consists of dimensionality reduction, Bayesian Network structure learning, conditional probability estimation, and probabilistic inference.

1. **Dimensionality Reduction Using ID3:** The ID3 algorithm is first used to evaluate the importance of each construction safety risk factor based on its Information Gain. Information Gain measures the reduction in uncertainty regarding the accident outcome obtained by considering a particular feature. Features with very low Information Gain provide limited predictive information and can therefore be removed from the model. A threshold value, represented by ϵ , is defined for feature pruning. If $\text{Gain}(X_i) < \epsilon$, where $\epsilon = 0.01$, the

corresponding feature is excluded from the Bayesian Network. Therefore, only features satisfying $\text{Gain}(X_i) \geq \epsilon$ are retained for further analysis. This dimensionality reduction process reduces the complexity of the network, improves computational efficiency, and ensures that the Bayesian Network focuses primarily on high-influence risk factors.

2. Bayesian Network Structure Learning and Conditional Probabilities: After feature selection, the remaining risk factors are represented as nodes in a Directed Acyclic Graph (DAG). Let the selected risk factors be represented as $X = \{X_1, X_2, \dots, X_k\}$, and let Y represent the construction safety accident outcome. The Bayesian Network models the conditional dependencies among these risk factors and the accident outcome. The joint probability distribution of all

$$P(X_1, X_2, \dots, X_k, Y) = P(Y | \text{Parents}(Y)) \times \prod_{i=1}^k P(X_i | \text{Parents}(X_i))$$

where $\text{Parents}(X_i)$ represents the set of parent nodes or direct causal predecessors of the risk factor X_i in the Directed Acyclic Graph. For example, a particular risk node may depend on factors such as inadequate training and the health condition of workers. Unlike models that assume complete independence between features, the Bayesian Network explicitly captures these conditional relationships. The Conditional Probability Tables (CPTs) associated with each node are estimated from the historical construction project database using Maximum Likelihood Estimation (MLE). Thus, the model learns the probability of each risk factor and accident outcome based on the observed frequencies and relationships in the available data.

3. Feasibility of the ID3-BN Model: The proposed extension is feasible because it uses the same historical frequency database already collected for the original research, consisting of 12 semi-annual records and 3,726 construction safety events. Therefore, no completely new dataset is required for the implementation of the model. The ID3 algorithm can first identify and remove low-influence features, after which the selected features can be used to construct the Bayesian Network. Software tools such as GeNIe, Netica, and Python libraries such as pgmpy can be used for Bayesian Network construction, structure learning, parameter estimation, and probabilistic inference. Since dimensionality reduction removes irrelevant features before the Bayesian Network is constructed, the computational complexity can also be reduced.

4. Expected Impact and Decision-Support Capability: The proposed ID3-BN model is expected to improve construction safety risk analysis in two important ways. First, it removes the strict assumption of feature independence by modelling the conditional relationships between different risk factors. This allows the model to evaluate risk-compounding scenarios, where multiple factors interact and collectively increase the probability of a serious accident. For example, minor technical faults combined with adverse weather conditions may create a significantly greater safety risk than either factor considered individually. Second, the Bayesian Network supports bidirectional probabilistic inference. Through forward inference, site managers can estimate the probability of a construction safety accident based on current site conditions and observed risk factors. Through backward or diagnostic inference, the model can analyze an observed accident and identify the most probable underlying causes or contributing risk factors. Consequently, the ID3-BN model provides a more comprehensive, dynamic, and powerful decision-support framework for construction safety risk prediction, diagnosis, and preventive management.